

Softmax

standard, 5-member ensemble, evaluated on 2016-2026, measured against the market

Contents

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|---|--|
| 1. Model identity | 7. Top-choice movement |
| 2. Executive summary | 8. Ensemble stability |
| 3. Evaluation population and market reference | 9. Feature and correction effects |
| 4. Overall probability performance | 10. Betting and bankroll analysis |
| 5. Year-by-year probability performance | 11. Simulation and uncertainty |
| 6. Calibration and sharpness | 12. Technical appendix and provenance |
| | 13. Limitations and the retrospective-odds warning |

Model identity

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Softmax standard	EVALUATION YEARS 11 2016-2026	ENSEMBLE MEMBERS 5 seeds 7, 17, 42, 123, 256	BOOSTING ROUNDS 400 3-fold chronological CV	FEATURE SET production 28 declared inputs
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Softmax is not the production model.

It is one of the standard published architectures, measured against the market on the same races as every other report here. Nothing in it is a promotion decision, and the promotion gates it can and cannot satisfy are in the technical appendix.

HOW EACH EVALUATION YEAR WAS FITTED

The 2016 evaluation model was trained on the initial 2010-2015 set. Each later evaluation adds the preceding complete season, so the training window expands by one year; the 2026 model uses 2010-2025. Every evaluation year is fitted separately -- 11 fits in all -- and no evaluation year's outcomes appear in its own fit.

Deltas in this report use `model_minus_market_log_loss: race_multinomial_log_loss`, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Provenance -- the source commit, the config and data digests, the environment and every input this report read by content digest -- is in the technical appendix and in `run_lock.json` beside this file. Canonical publication fails without it.

Executive summary

MEAN DELTA VS MARKET -0.000992 negative is better	RACES 7,933	YEARS BETTER 7/11	RACE-LEVEL WIN RATE 51.7%
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Softmax is better than the market by 0.000992 `race_multinomial_log_loss` on 7,933 races. The direction holds in 7 of 11 evaluated years. Race by race it is ahead 51.7% of the time, which is the more honest figure: a mean delta can be carried by a few races. 13,793 bets were selected under the staking rules below, on retrospective odds that could not have been bet into.

Evaluation population and market reference

THE MARKET REFERENCE

Every metric below is reported beside the market's own implied probabilities on the same races, normalised within each race, and the delta is this model's race log loss minus the market's on the same races. One yardstick across every published report, so their headline numbers mean the same thing and can be read against one another. The market is a measurement, not a model this one is defined against.

THE EVALUATION POPULATION

Every figure in this report covers the standardized evaluation window, which begins in 2016. One window for every published report, so their headline numbers are computed on the identical race keys and can be compared -- which is checked by digest before anything is published, not asserted in a footnote. 2015 is earlier than the window and is excluded from every evaluation figure below, including the bankroll paths. The feature-effect grid is the one exception, and says so where it appears: a learned response direction is a property of each year's fit rather than of the evaluation population.

TEST RACES 7,933	YEARS 11 2016-2026
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Splits present: test, train_apparent. Only the forward test split is scored above. The other exists to be compared against it -- an apparent-fit number looks far better than a forward one -- and is never pooled with it.

Out-of-fold margin sources: model.

One or more years is partial

2026 does not span a full season, so its per-year figures rest on fewer races than the others. Partial years are marked in every table rather than dropped, because dropping them would silently change the evaluation population.

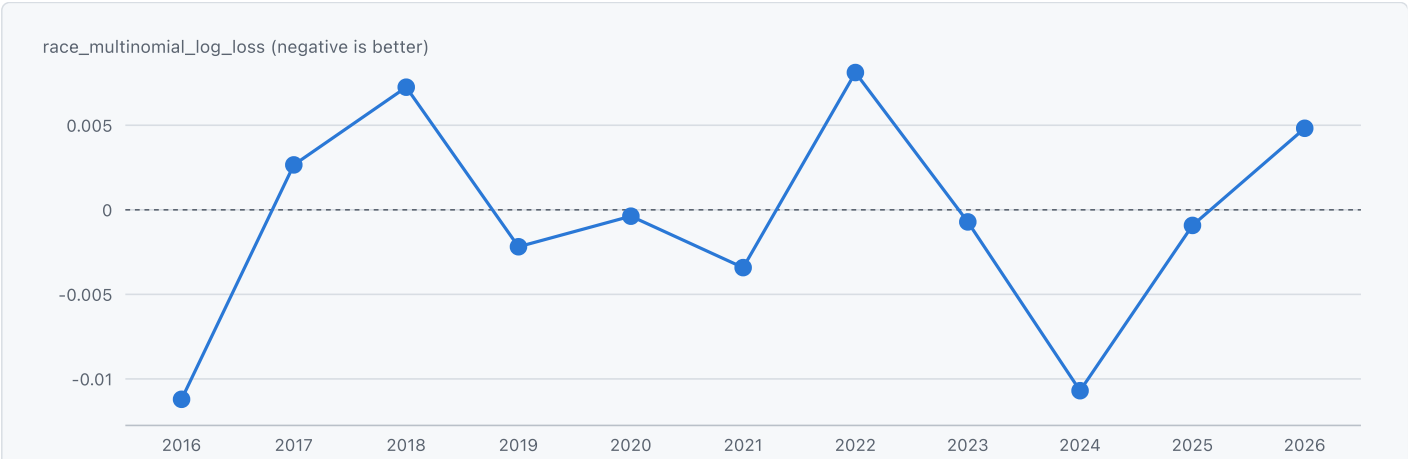
Overall probability performance

races	model log loss	market log loss	model minus market log loss	better race share
7933	2.027671	2.028662	-0.000992	0.517459

One row, pooled over every evaluated race, and measured against the market: model_minus_market_log_loss is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	model minus market log loss	better race share	is partial year
2016	744	2.088076	2.099277	-0.011201	0.534946	no
2017	772	2.024210	2.021559	0.002650	0.503886	no
2018	756	2.068746	2.061504	0.007242	0.489418	no
2019	759	1.930639	1.932827	-0.002188	0.524374	no
2020	799	2.010022	2.010400	-0.000378	0.508135	no
2021	783	2.003245	2.006659	-0.003414	0.530013	no
2022	783	2.059305	2.051195	0.008111	0.507024	no
2023	796	2.023345	2.024066	-0.000721	0.520101	no
2024	792	2.015049	2.025744	-0.010695	0.542929	no
2025	748	2.056470	2.057397	-0.000927	0.521390	no
2026	201	2.031045	2.026224	0.004821	0.487562	yes



Mean delta by year Mean delta against the market for each evaluated year. Zero is the market.

Table view

Category	race_multinomial_log_loss (negative is better)
2016	-0.011201
2017	0.00265
2018	0.007242
2019	-0.002188
2020	-0.000378
2021	-0.003414
2022	0.008111
2023	-0.000721

Category	race_multinomial_log_loss (negative is better)
2024	-0.010695
2025	-0.000927
2026	0.004821

Ahead of the market in 7 of 11 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

Reliability by pipeline stage, on bins fixed across the whole report so two stages are comparable, pooled from the per-year counts rather than averaged over years. Stages: market, raw_utility, temperature, bagged.

MARKET

Pooled over every test year.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	10,889	36	0.0062	0.0033	0.538
0.01-0.03	16,603	263	0.0171	0.0158	0.925
0.03-0.05	17,886	625	0.0365	0.0349	0.956
0.05-0.07	15,094	901	0.0625	0.0597	0.955
0.07-0.10	10,318	843	0.0874	0.0817	0.935
0.10-0.15	11,807	1,511	0.1223	0.1280	1.046
0.15-0.20	6,323	1,130	0.1728	0.1787	1.034
0.20-0.30	5,668	1,389	0.2419	0.2451	1.013
0.30-0.50	2,723	1,040	0.3651	0.3819	1.046
0.50-1.00	325	195	0.5566	0.6000	1.078

RAW_UTILITY

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	64,903	287	0.0047	0.0044	0.944
0.01-0.03	80,418	1,407	0.0168	0.0175	1.039
0.03-0.05	90,429	3,323	0.0383	0.0367	0.959
0.05-0.07	79,377	4,950	0.0616	0.0624	1.013
0.07-0.10	43,299	3,785	0.0863	0.0874	1.013
0.10-0.15	48,670	6,359	0.1229	0.1307	1.063
0.15-0.20	33,185	5,683	0.1726	0.1713	0.992
0.20-0.30	30,535	7,159	0.2427	0.2345	0.966
0.30-0.50	15,304	5,528	0.3697	0.3612	0.977
0.50-1.00	2,060	1,184	0.5548	0.5748	1.036

TEMPERATURE

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

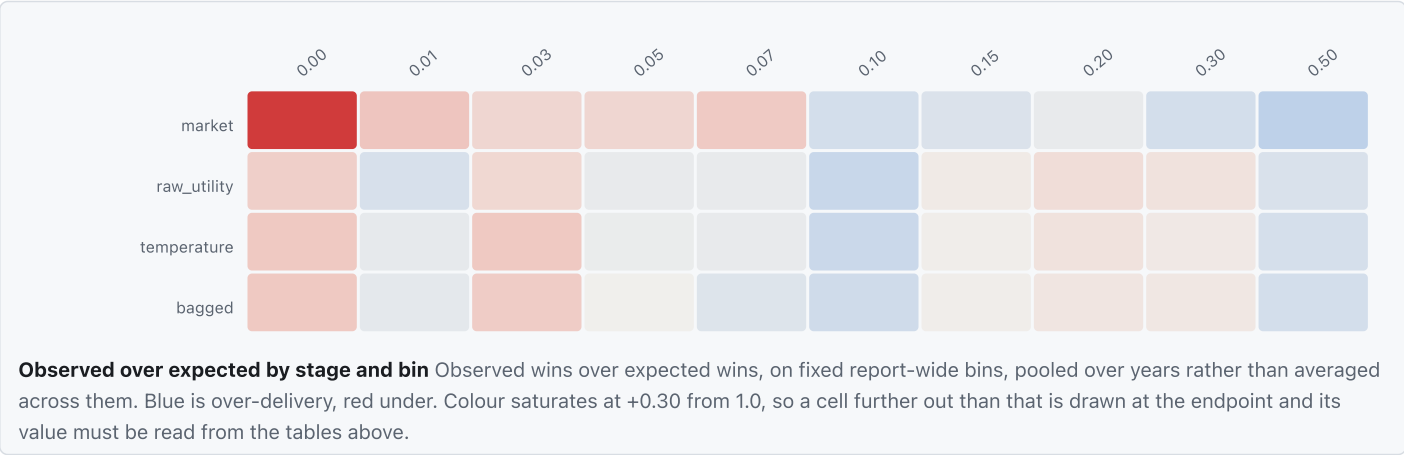
bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	63,470	281	0.0048	0.0044	0.932
0.01-0.03	80,578	1,390	0.0170	0.0173	1.015
0.03-0.05	89,431	3,199	0.0384	0.0358	0.932
0.05-0.07	80,608	5,013	0.0616	0.0622	1.009
0.07-0.10	44,071	3,853	0.0863	0.0874	1.013
0.10-0.15	49,335	6,423	0.1229	0.1302	1.060
0.15-0.20	33,354	5,732	0.1725	0.1719	0.996
0.20-0.30	30,467	7,201	0.2424	0.2364	0.975
0.30-0.50	14,923	5,448	0.3695	0.3651	0.988

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.50-1.00	1,943	1,125	0.5555	0.5790	1.042

BAGGED

The bagged probabilities, pooled over every test year.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	12,665	56	0.0047	0.0044	0.932
0.01-0.03	16,176	280	0.0170	0.0173	1.018
0.03-0.05	17,875	643	0.0384	0.0360	0.936
0.05-0.07	16,112	994	0.0616	0.0617	1.001
0.07-0.10	8,799	781	0.0863	0.0888	1.029
0.10-0.15	9,864	1,272	0.1229	0.1290	1.050
0.15-0.20	6,684	1,149	0.1725	0.1719	0.997
0.20-0.30	6,120	1,457	0.2427	0.2381	0.981
0.30-0.50	2,953	1,076	0.3700	0.3644	0.985
0.50-1.00	388	225	0.5550	0.5799	1.045



The largest departure is the 0.000 to 0.010 bin, and every stage misses there: market 0.54, raw_utility 0.94, temperature 0.93, bagged 0.93. market is furthest from 1.0 at 0.54, meaning that bin won fewer races than it priced; raw_utility is closest at 0.94, so this bin is where the pipeline moves calibration most. On the market stage that bin holds 10,889 runners, the 6th smallest of 10 bins, so the departure is not a thin-bin artefact. Calibration in this bin does not improve monotonically through the pipeline: temperature reaches 0.93 and bagged falls back to 0.93, so that step gives back part of the correction rather than adding to it.

SHARPNESS

stage	racess	mean entropy	mean normalised entropy	mean top probability	mean field size
market	7,933	2.0688	0.8274	0.2907	12.3076
raw_utility	7,933	2.0242	0.8099	0.3075	12.3076
temperature	7,933	2.0321	0.8130	0.3047	12.3076
bagged	7,933	2.0325	0.8132	0.3045	12.3076

Entropy normalised by field size, so an eight-runner race and a fourteen-runner race are on one scale, and averaged over years weighted by races rather than as a mean of yearly means. A sharper distribution is not automatically better: it is only better if the calibration above holds.

Top-choice movement

test year	n races	n disagreement races	disagreement share	model top win rate	market top win rate	disagreement mean delta	agreement mean delta	seed top choice agreement
2016	744	93	0.125000	0.236559	0.258065	-0.016243	-0.010481	0.894624
2017	772	72	0.093264	0.333333	0.180556	-0.019840	0.004964	0.866667
2018	756	90	0.119048	0.211111	0.122222	0.000588	0.008141	0.911111
2019	759	60	0.079051	0.233333	0.233333	-0.004541	-0.001986	0.886667
2020	799	62	0.077597	0.161290	0.209677	0.040305	-0.003801	0.870968
2021	783	79	0.100894	0.215190	0.240506	0.005022	-0.004361	0.896203
2022	783	58	0.074074	0.172414	0.258621	-0.011525	0.009681	0.879310

test year	n races	n disagreement races	disagreement share	model top win rate	market top win rate	disagreement mean delta	agreement mean delta	seed top choice agreement
2023	796	49	0.061558	0.061224	0.326531	0.017677	-0.001928	0.889796
2024	792	61	0.077020	0.163934	0.213115	0.026125	-0.013768	0.881967
2025	748	63	0.084225	0.142857	0.174603	-0.018019	0.000645	0.907937
2026	201	16	0.079602	0.250000	0.250000	0.019545	0.003547	0.875000

How often this model's first pick differs from the market's, and whether it is right when it does. Disagreeing more is not better on its own -- the useful figure is the win rate *conditional* on disagreeing, because a model that disagrees often and loses is worse than one that never disagrees. The last column is how often the seeds pick the same runner as each other, which bounds how much of any disagreement is signal rather than seed noise.

Ensemble stability

RUNNERS COVERED
97,636

YEARS
11

SEEDS BAGGED
5

Member dispersion per runner. The bag is only worth its cost if members genuinely disagree: an ensemble whose members agree everywhere is one fit with five times the runtime. The per-runner spread across members is in `ensemble_diagnostics.parquet`, race by race.

Feature and correction effects

These are contributions to the whole utility, measured from the booster's intercept

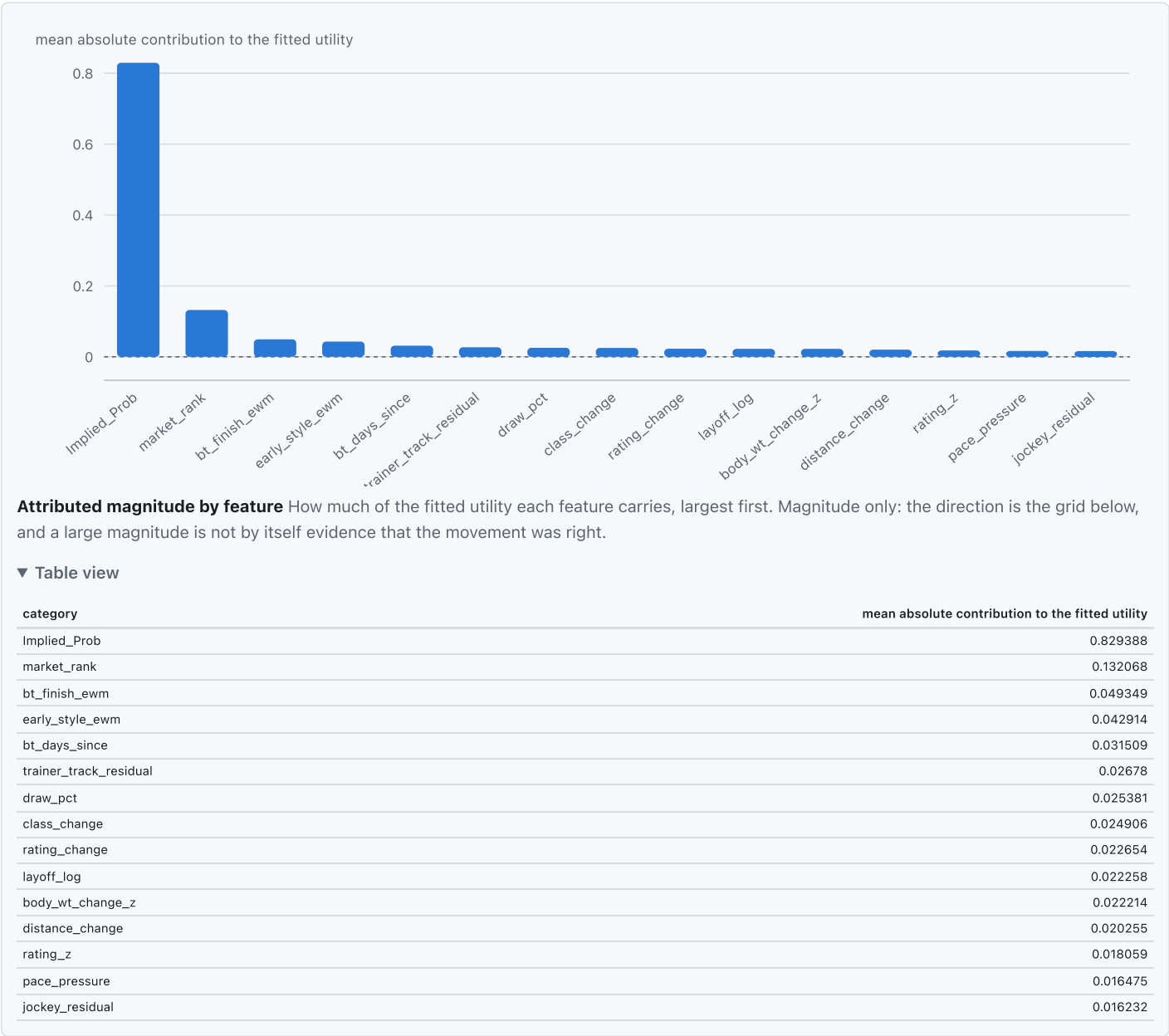
Each feature's effect below is its contribution to this model's whole fitted utility, not to a correction of the market. The baseline is the booster's own intercept, and the market's implied probability is one of the features being attributed rather than the point of comparison -- so these magnitudes are larger than the market-anchored reports' and are not comparable with them term by term.

2 features cannot be tested this way.

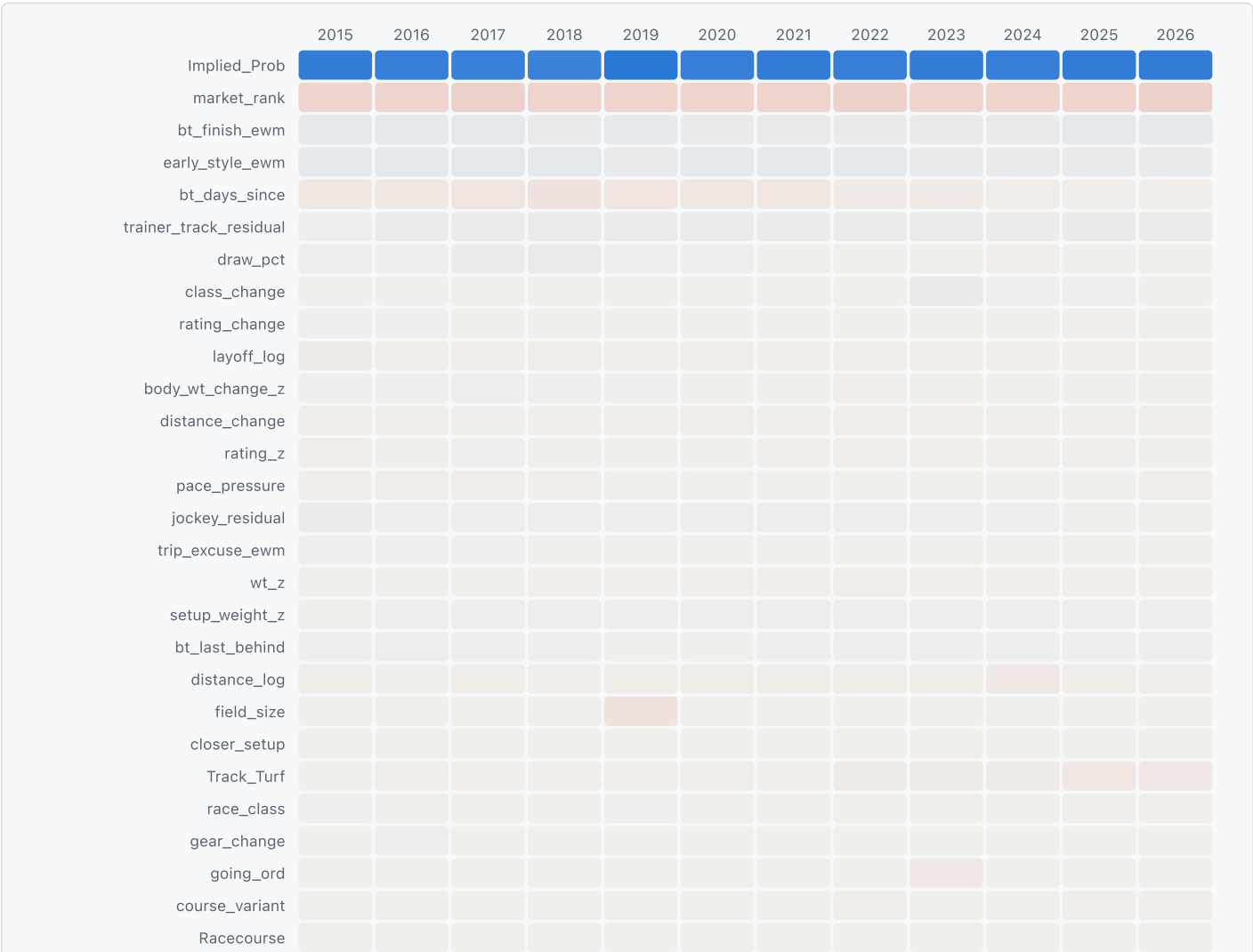
Implied_Prob, market_rank belong to no group, so no ablation can remove them and this method cannot measure their held-out value.

label	feature	mean abs attributed effect	sign agreement	rank stability	seed variation	psi	practical support	feature type	n years
stable_positive	Implied_Prob	0.829388	1	1	0.053694	0.032185	12	continuous	12
stable_nonlinear	market_rank	0.132068	1	1	0.341955	0.001269	12	continuous	12
stable_positive	bt_finish_ewm	0.049349	0.975000	0.599320	0.093379	0.046310	12	continuous	12
stable_positive	early_style_ewm	0.042914	0.983333	0.599320	0.089325	0.007011	12	continuous	12
stable_nonlinear	bt_days_since	0.031509	0.916667	0.244598	0.110599	0.093612	12	continuous	12
stable_positive	trainer_track_residual	0.026780	0.941667	0.213656	0.093600	0.129261	12	continuous	12
stable_nonlinear	draw_pct	0.025381	0.941667	0.292672	0.134900	0.000533	12	continuous	12
stable_nonlinear	class_change	0.024906	0.783333	0.389973	0.277523	0.006694	12	categorical	12
stable_nonlinear	rating_change	0.022654	0.966667	0.288353	0.150650	0.045097	12	continuous	12
stable_nonlinear	layoff_log	0.022258	0.941667	0.286460	0.179465	0.103384	12	continuous	12
stable_nonlinear	body_wt_change_z	0.022214	0.933333	0.224726	0.139412	0.010553	12	continuous	12
stable_nonlinear	distance_change	0.020255	0.958333	0.211485	0.184851	0.019978	12	continuous	12
stable_nonlinear	rating_z	0.018059	0.925000	0.294709	0.395350	0.021924	12	continuous	12
stable_nonlinear	pace_pressure	0.016475	0.885417	0.162079	0.137859	0.064975	12	continuous	12
distribution_shifted	jockey_residual	0.016232	0.908333	0.204273	0.217952	0.794525	12	continuous	12
stable_nonlinear	trip_excuse_ewm	0.015905	0.833333	0.226240	0.122831	0.058036	12	continuous	12
stable_nonlinear	wt_z	0.015313	0.958333	0.288865	0.214655	0.035055	12	continuous	12
stable_nonlinear	setup_weight_z	0.013560	0.858333	0.373624	0.327083	0.002993	12	continuous	12
stable_nonlinear	bt_last_behind	0.012400	0.824074	0.169277	0.199479	0.026302	12	continuous	12
stable_nonlinear	distance_log	0.012259	1	0.336109	0.089951	0.072032	12	categorical	12
stable_nonlinear	field_size	0.010445	0.833333	0.327815	0.393596	0.106798	12	categorical	12
stable_nonlinear	closer_setup	0.007817	0.800000	0.432643	0.204509	0.057756	12	continuous	12
stable_negative	Track_Turf	0.003677	1	0.314520	0.306525	0.004738	12	binary	12

label	feature	mean abs attributed effect	sign agreement	rank stability	seed variation	psi	practical support	feature type	n years
stable_nonlinear	race_class	0.003234	1	0.616682	0.413877	0.055002	12	categorical	12
regime_dependent	gear_change	0.003223	0.583333	0.294847	0.209080	0.018293	8	binary	12
seed_unstable	going_ord	0.002683	0.591667	0.373624	0.645532	0.355145	12	categorical	12
stable_nonlinear	course_variant	0.002015	1	0.605569	0.372707	0.008740	12	categorical	12
stable_negative	Racecourse	0.001001	1	0.597300	0.136758	0.002257	12	binary	12



DIRECTION BY FEATURE AND YEAR



Response direction by feature and year Each cell is that year's tail effect: the last response bin minus the first, averaged over seeds. Blue is a response that rises across the feature's range and red one that falls; a row of one colour is a feature whose direction held every year. Colour extent is per grid, so this shows relative consistency rather than absolute size -- the sizes are the bar chart above.

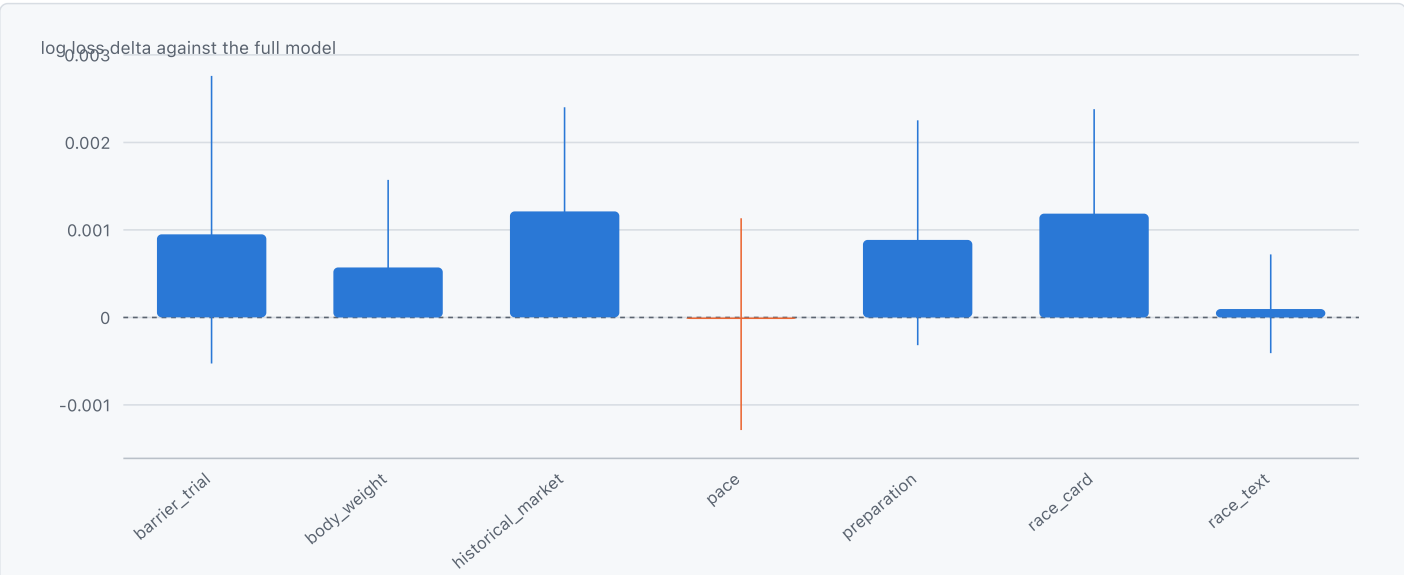
feature	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Implied_Prob	0.235053	0.233050	0.228730	0.226891	0.244370	0.233195	0.235965	0.233219	0.233763	0.232194	0.237616	0.236509
market_rank	-0.038047	-0.040489	-0.043755	-0.041474	-0.041233	-0.041001	-0.041434	-0.043320	-0.042537	-0.042050	-0.042020	-0.043122
bt_finish_ewm	0.012034	0.012052	0.011332	0.009149	0.011406	0.009148	0.010976	0.010987	0.010804	0.010493	0.011441	0.012637
early_style_ewm	0.012826	0.013777	0.013961	0.011891	0.010823	0.011219	0.012585	0.011330	0.010585	0.010888	0.010793	0.010779
bt_days_since	-0.011114	-0.013080	-0.014915	-0.016821	-0.014283	-0.011421	-0.011182	-0.008474	-0.006777	-0.004449	-0.003233	-0.003474
trainer_track_residual	0.002537	0.007682	0.006743	0.007538	0.006869	0.007068	0.007525	0.007030	0.006944	0.008303	0.009298	0.008365
draw_pct	0.004882	0.003531	0.006901	0.006206	0.002768	0.002487	-0.000050	-0.000619	-0.001186	-0.001422	0.000440	0.000410
class_change	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	0.008950	0.002260	0.003309	n/a
rating_change	0.002054	0.001373	0.000138	0.000463	0.000762	0.000184	-0.000235	-0.000903	-0.000555	-0.000209	-0.002524	-0.000891
layoff_log	-0.007036	-0.005059	-0.005527	-0.005131	-0.005423	-0.004627	-0.002796	-0.003334	-0.003230	-0.002896	-0.002993	-0.003660
body_wt_change_z	0.003157	0.002284	0.003750	0.003462	0.002744	0.000417	0.000095	-0.001415	-0.000582	-0.000416	-0.000002	0.000085
distance_change	-0.004420	-0.003199	-0.002563	-0.003500	-0.004196	-0.004176	-0.003426	-0.003276	-0.003741	-0.002880	-0.002378	-0.001839
rating_z	-0.004045	-0.000767	0.001433	0.001207	0.000207	-0.000491	-0.001319	-0.001397	-0.002080	-0.002990	-0.001892	-0.000725
pace_pressure	-0.003338	-0.004956	-0.003842	-0.001963	0.000127	0.000164	-0.001036	-0.000915	-0.001074	-0.001187	-0.000938	-0.001342
jockey_residual	0.008158	0.005831	0.003737	0.004631	0.003180	0.003671	0.005567	0.005384	0.005084	0.001503	0.001213	0.000698
trip_excuse_ewm	0.002497	0.001372	0.001356	0.000574	-0.001045	-0.001172	-0.000219	0.000181	-0.000618	0.000659	0.001094	0.001115
wt_z	0.000509	0.000514	-0.000006	-0.002455	-0.002186	-0.003265	-0.002811	-0.003837	-0.002206	-0.002794	-0.002419	-0.002524
setup_weight_z	0.000368	0.003054	0.004746	0.005598	0.004759	0.003962	0.004488	0.003724	0.003670	0.001934	0.002582	0.002600
bt_last_behind	0.003146	0.004365	0.003524	0.001315	0.000950	0.001133	0.001586	0.001922	0.001805	0.001314	0.001366	0.001440
distance_log	-0.004890	0.000455	-0.003907	-0.001346	-0.005862	-0.005385	-0.004838	-0.004337	-0.005170	-0.008632	-0.005975	-0.003617
field_size	n/a	n/a	n/a	n/a	-0.016253	n/a	n/a	n/a	n/a	n/a	n/a	n/a

feature	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
closer_setup	-0.000694	-0.000530	0.000436	-0.000012	0.000488	-0.000148	0.000174	0.000352	0.000428	0.000915	0.000911	0.000458
Track_Turf	-0.001068	-0.002036	-0.002840	-0.001741	-0.005687	-0.004662	-0.003109	-0.007979	-0.007965	-0.007949	-0.011860	-0.009601
race_class	0.001849	0.001025	0.000257	-0.000385	0.001223	-0.000310	0.000154	0.000577	0.000306	-0.000774	-0.001993	-0.002506
gear_change	0.000638	0.001266	0.001030	0.000664	0.000340	0.000087	0.000044	-0.000112	-0.000266	-0.000213	-0.000144	-0.000306
going_ord	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	-0.010756	n/a	n/a	n/a
course_variant	0.000878	0.000518	-0.000032	-0.000107	-0.000724	-0.000903	0.000169	-0.001892	-0.001059	-0.001202	-0.001626	-0.001753
Racecourse	-0.000280	-0.000364	-0.000396	-0.000708	-0.000372	-0.000431	-0.000650	-0.000577	-0.000757	-0.000500	-0.000533	-0.000546

This grid covers every year the run fitted, including any the evaluation population excludes. A learned response direction is a property of that year's fit rather than of the race population the headline metrics are computed on, and its per-feature mean is the stability table's tail_effect exactly -- so restricting one and not the other would have two parts of this report describe different populations. 14 of 28 features keep the same sign in every year they appear in. For the seven features that are constant within a race -- a racecourse, a surface -- the cell is the correction's magnitude and not its direction, because a race-constant feature cancels from every pairwise utility difference and has no direction to report. Which basis a row uses is in its tooltip and in the stability table's curve_basis column. A feature whose row changes colour is not necessarily broken: it may genuinely matter differently in different years. But its mean effect is then a summary of disagreeing years, and the mean is the least informative number about it.

GROUP ABLATION

group	mean delta vs full	bootstrap lower	bootstrap upper	n years	n seeds
barrier_trial	0.000949	-0.000526	0.002760	12	5
body_weight	0.000572	0.000265	0.001571	12	5
historical_market	0.001211	0.000376	0.002401	12	5
pace	-0.000016	-0.001287	0.001134	12	5
preparation	0.000885	-0.000317	0.002253	12	5
race_card	0.001187	0.000239	0.002380	12	5
race_text	0.000097	-0.000408	0.000719	12	5



Held-out value by feature group Each group removed and the model refitted, scored held out, with its bootstrap interval. Positive means removing the group made the model worse, so the group was carrying value. A bar whose interval spans zero was not distinguishable from noise.

▼ Table view

category	log loss delta against the full model	bootstrap interval
barrier_trial	0.000949	-0.000526 to 0.00276
body_weight	0.000572	0.000265 to 0.001571
historical_market	0.001211	0.000376 to 0.002401
pace	-0.000016	-0.001287 to 0.001134
preparation	0.000885	-0.000317 to 0.002253
race_card	0.001187	0.000239 to 0.00238
race_text	0.000097	-0.000408 to 0.000719

The deltas do *not* sum to the whole: features share information, so attributing a total by adding these up overstates it.

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

Softmax is not the production model, and this is not a proposal to stake it.

A settled ledger measures what a probability was worth economically, which is a different question from which model should carry money. That decision is the production model's, and the comparison that would inform it is in the technical appendix rather than in this table.

SETTLEMENT BY STAKING RULE

Role	What it is for
Primary	what an operator would run: sized for the rebate, and credited it.
Sizing reference	the same settlement, sized blind. It isolates sizing for the cliff from receiving the rebate -- this row and the primary settle identically and differ only in what the solver knew.
Settlement ablation	blind sizing with the rebate not credited. It isolates the rebate's own contribution to the return.
Benchmark	a fixed flat stake above an edge floor, sized from the opening bankroll and never resized. Not a Kelly fraction, so it belongs to no preset panel below -- and it has no ruin stop, so a model that finds edge in most races can stake more than the bankroll holds and finish below zero. That is the yardstick behaving as defined rather than a settlement error: it is here to show what indiscriminate staking on this model's edges would have done, which is why it is a benchmark and not a proposal.

The table below is ordered by that role rather than alphabetically, so the first Kelly rows are the ones an operator would run.

Flat stake@edge>=0.02 closed below zero.

The flat-stake benchmark is sized from the opening bankroll and never resized, so it has no ruin stop: a model that finds edge in most races stakes more than the bankroll holds. The closing figure is what that rule would have owed, not a bankroll that existed. Every Kelly rule here sizes from the running bankroll and cannot do this.

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
rebate_aware_kelly@0.01	10,632	12,820,920	-988715.500000	-0.077117	-0.047639	0.087615	9534776.500000	5873
rebate_aware_kelly@0.02	11,089	38,989,020	-2542657.500000	-0.065215	0.011014	0.116248	10110748.450000	6013
rebate_aware_kelly@0.05	12,153	123,552,640	-8,134,516	-0.065838	0.140166	0.250029	11504652.250000	6642
rebate_aware_kelly@0.1	12,716	251,742,430	-20,445,974	-0.081218	-0.057856	0.511237	9437860.100000	7131
kelly_settled_rebate@0.01	9927	8,008,280	-521788.500000	-0.065156	-0.051693	0.072543	9496202.500000	5880
kelly_settled_rebate@0.02	9965	16,112,800	-1051991.500000	-0.065289	-0.083431	0.128438	9199543.500000	5896
kelly_settled_rebate@0.05	9980	42,932,450	-2908646.500000	-0.067749	-0.080535	0.222943	9226227.500000	5903
kelly_settled_rebate@0.1	9982	96,902,060	-7031328.500000	-0.072561	-0.029139	0.339203	9712813.500000	5902
kelly_no_rebate@0.01	9927	7,998,720	-520,545	-0.065079	-0.053458	0.073472	9,479,455	5880
kelly_no_rebate@0.02	9965	15,878,180	-1,024,910	-0.064548	-0.108132	0.142216	8,975,090	5904
kelly_no_rebate@0.05	9978	38,759,830	-2,437,268	-0.062881	-0.279353	0.322574	7,562,732	5914
kelly_no_rebate@0.1	9980	74,144,960	-4,450,311	-0.060022	-0.588843	0.549760	5,549,689	5917
flat_stake@edge>=0.02	8364	83,640,000	-12,074,000	-0.144357	-36.841361	1.194001	-2,074,000	5263

Model view softmax_bagged. Seed -1. Bankroll scenario hkd_10m, opening HK\$10,000,000 (primary). These rules see the same races and the same odds. They do not necessarily choose the same runners or the same races: rebate-aware sizing admits runners a blind solve would reject,

because staking a mildly negative-edge runner raises the loss on every other outcome and can push a race over the rebate threshold. Compare the bet counts as well as the returns.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

Bankroll scenario hkd_10m, opening HK\$10,000,000 (primary).

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.034210	0.034210	-0.1923	1856
0.050000 to 0.100000	0.072648	0.072648	-0.1929	2224
0.100000 to 0.200000	0.140811	0.140811	0.0967	2369
0.200000 to unbounded	0.352988	0.352988	-0.3397	1915

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.010208	0.010208	-0.0766	-0.0757	-0.0738	-0.0681	1563	1601	1614	1616
0.020000 to 0.050000	0.034210	0.034210	-0.0214	-0.0201	-0.0168	-0.0100	1856	1856	1856	1856
0.050000 to 0.100000	0.072648	0.072648	-0.1658	-0.1664	-0.1681	-0.1709	2224	2224	2224	2224
0.100000 to 0.200000	0.140811	0.140811	-0.0048	-0.0045	-0.0028	-0.0011	2369	2369	2369	2369
0.200000 to unbounded	0.352988	0.352988	-0.0725	-0.0713	-0.0682	-0.0623	1915	1915	1915	1915

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.010208	0.010208	-0.0764	-0.0770	-0.0751	-0.0699	1563	1601	1616	1618
0.020000 to 0.050000	0.034210	0.034210	-0.0215	-0.0211	-0.0238	-0.0288	1856	1856	1856	1856
0.050000 to 0.100000	0.072648	0.072648	-0.1659	-0.1664	-0.1685	-0.1716	2224	2224	2224	2224
0.100000 to 0.200000	0.140811	0.140811	-0.0046	-0.0047	-0.0052	-0.0070	2369	2369	2369	2369
0.200000 to unbounded	0.352988	0.352988	-0.0729	-0.0730	-0.0786	-0.0894	1915	1915	1915	1915

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.018875	-0.018875	-0.1466	-0.0356	-0.0894	-0.1382	703	1122	2193	2754
0 to 0.020000	0.010207	0.010207	-0.0727	-0.0943	-0.0375	-0.0279	1565	1603	1596	1598
0.020000 to 0.050000	0.034210	0.034210	-0.0560	0.0143	-0.0254	-0.0521	1856	1856	1856	1856
0.050000 to 0.100000	0.072648	0.072648	-0.1828	-0.1272	-0.1566	-0.1726	2224	2224	2224	2224
0.100000 to 0.200000	0.140811	0.140811	-0.0237	-0.0419	-0.0003	-0.0149	2369	2369	2369	2369
0.200000 to unbounded	0.352988	0.352988	-0.0399	-0.0902	-0.0836	-0.0790	1915	1915	1915	1915

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL

HK\$10,000,000

scenario hkd_10m

PRESETS SETTLED

1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly

PATH MODES

annual reset, continuous

SEEDS

5

7, 17, 42, 123, 256

TRAINED ON

initial 2010-2015,

expanding by one season per evaluation year

a separate fit per evaluation year, each on 2010 through the year before it -- 11 fits in all

BOOSTING ROUNDS

400

YEARS STAKED

2016-2026

2015 excluded

BANKROLL SCENARIOS

2

hkd_10m, hkd_1b; each solved independently

REBATE THRESHOLD

1e-03 of the bank

a fixed HK\$10,000, so its bankroll fraction is what makes two scenarios differ

The 2016 evaluation model was trained on the initial 2010-2015 set. Each later evaluation adds the preceding complete season, so the training window expands by one year; the 2026 model uses 2010-2025. Every evaluation year is fitted separately -- 11 fits in all -- and no evaluation year's outcomes appear in its own fit. Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. These curves are drawn from a bounded 1,000-point subsequence of up to 7,529 staked races, keeping each interval's highest and lowest balance so no peak or drawdown is smoothed away, and keeping the first and last points exactly. Every figure in the tables is computed from the full settlement, and the complete per-race path is in `race_paths.parquet`. Where a year is excluded, a continuous path opens at the balance the bank held when the kept years begin and an annual-reset path opens at the declared bankroll, so the two show different opening figures for the same rule. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. annual-reset re-settlement of continuous-path decisions: the same stakes, compounded from the declared opening balance at the start of each test year. Not an annual-reset sizing simulation -- stakes are not re-solved from reset capital, and doing so would change which bets are placed rather than only how they compound. Each bankroll scenario below is solved from its own opening balance, sequentially by race, because the rebate threshold is a fixed dollar amount and the fraction the solver sees therefore changes with the bank. The larger scenario is not the smaller one's stakes scaled up. 2015 is excluded from every path below. That year's utility scale fell back to its theoretical value rather than being fitted, so its stakes are not comparable with the rest.

HK\$10.1M, 1% KELLY

OPENING BANKROLL

HK\$10,113,498

scenario hkd_10m, solved independently

PRIMARY FINAL RETURN

-5.7%

Rebate-aware Kelly, continuous

BETS

9,279

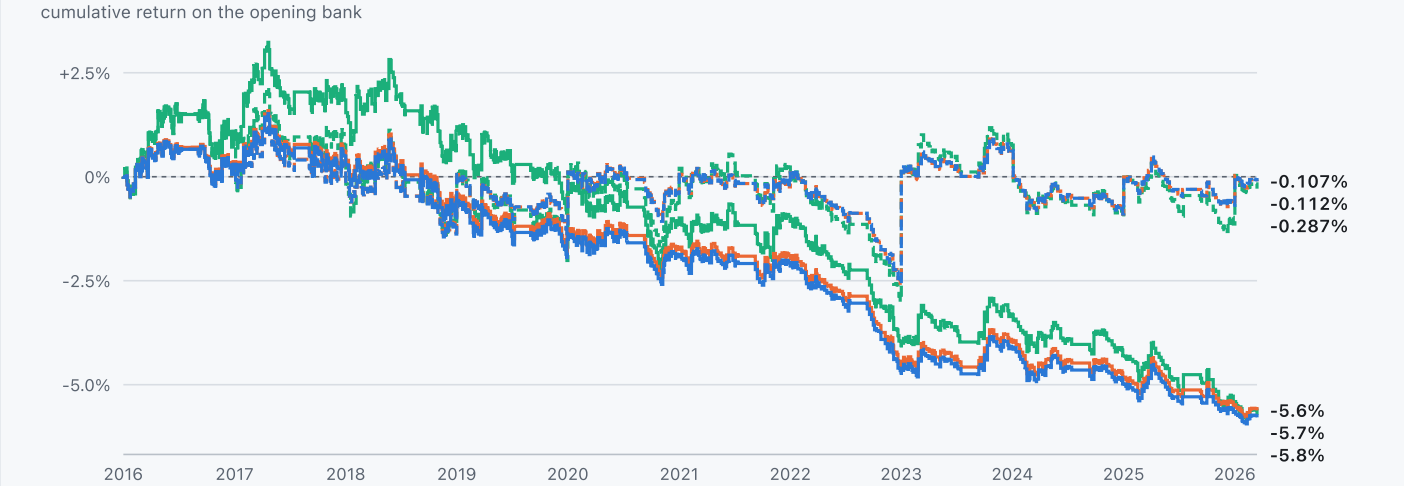
settled bets

REBATE RECEIVED

HK\$454,836

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	998	-0.287%	9,279 bets
Rebate-aware Kelly, continuous	998	-5.7%	9,279 bets
Kelly, settled rebate, annual reset	998	-0.107%	8,695 bets
Kelly, settled rebate, continuous	998	-5.6%	8,695 bets
Kelly, no rebate, annual reset	998	-0.112%	8,695 bets
Kelly, no rebate, continuous	998	-5.8%	8,695 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	1,197	1,101	1,101
2017	1,143	1,054	1,054
2018	1,059	970	970
2019	1,095	1,017	1,017
2020	1,108	1,042	1,042
2021	783	735	735
2022	780	749	749
2023	651	620	620
2024	684	661	661
2025	626	602	602
2026	153	144	144

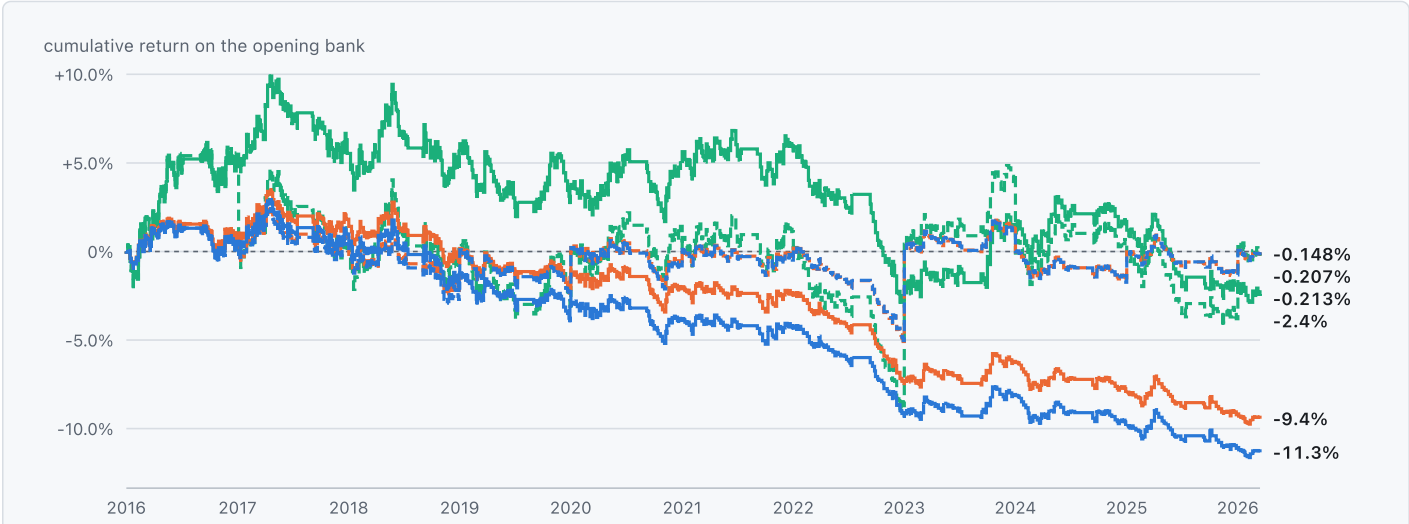
2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 1,197 bets in 2016 against 626 in 2025. Kelly, settled rebate and Kelly, no rebate: 1,101 bets in 2016 against 602 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$454,836; Rebate-aware Kelly, continuous HK\$454,836; Kelly, settled rebate, annual reset HK\$17,991; Kelly, settled rebate, continuous HK\$17,991. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

HK\$10.4M, 2% KELLY

OPENING BANKROLL HK\$10,361,374 scenario hkd_10m, solved independently	PRIMARY FINAL RETURN -2.4% Rebate-aware Kelly, continuous	BETS 9,717 settled bets	REBATE RECEIVED HK\$2,329,418 by the primary rule
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Rebate-aware Kelly
Kelly, settled rebate
Kelly, no rebate



Cumulative bankroll return at 2% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	999	-0.148%	9,717 bets
Rebate-aware Kelly, continuous	999	-2.4%	9,717 bets
Kelly, settled rebate, annual reset	998	-0.213%	8,730 bets
Kelly, settled rebate, continuous	998	-9.4%	8,730 bets
Kelly, no rebate, annual reset	998	-0.207%	8,730 bets
Kelly, no rebate, continuous	998	-11.3%	8,730 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	1,245	1,106	1,106
2017	1,197	1,058	1,058
2018	1,095	972	972
2019	1,134	1,022	1,022
2020	1,154	1,047	1,047
2021	841	742	742
2022	818	751	751
2023	686	621	621
2024	723	662	662
2025	658	604	604
2026	166	145	145

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 1,245 bets in 2016 against 658 in 2025. Kelly, settled rebate and Kelly, no rebate: 1,106 bets in 2016 against 604 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$2,329,418; Rebate-aware Kelly, continuous HK\$2,329,418; Kelly, settled rebate, annual reset HK\$215,905; Kelly, settled rebate, continuous HK\$215,905. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

HK\$10.6M, 5% KELLY

OPENING BANKROLL

HK\$10,573,811

scenario hkd_10m, solved independently

PRIMARY FINAL RETURN

+8.8%

Rebate-aware Kelly, continuous

BETS

10,712

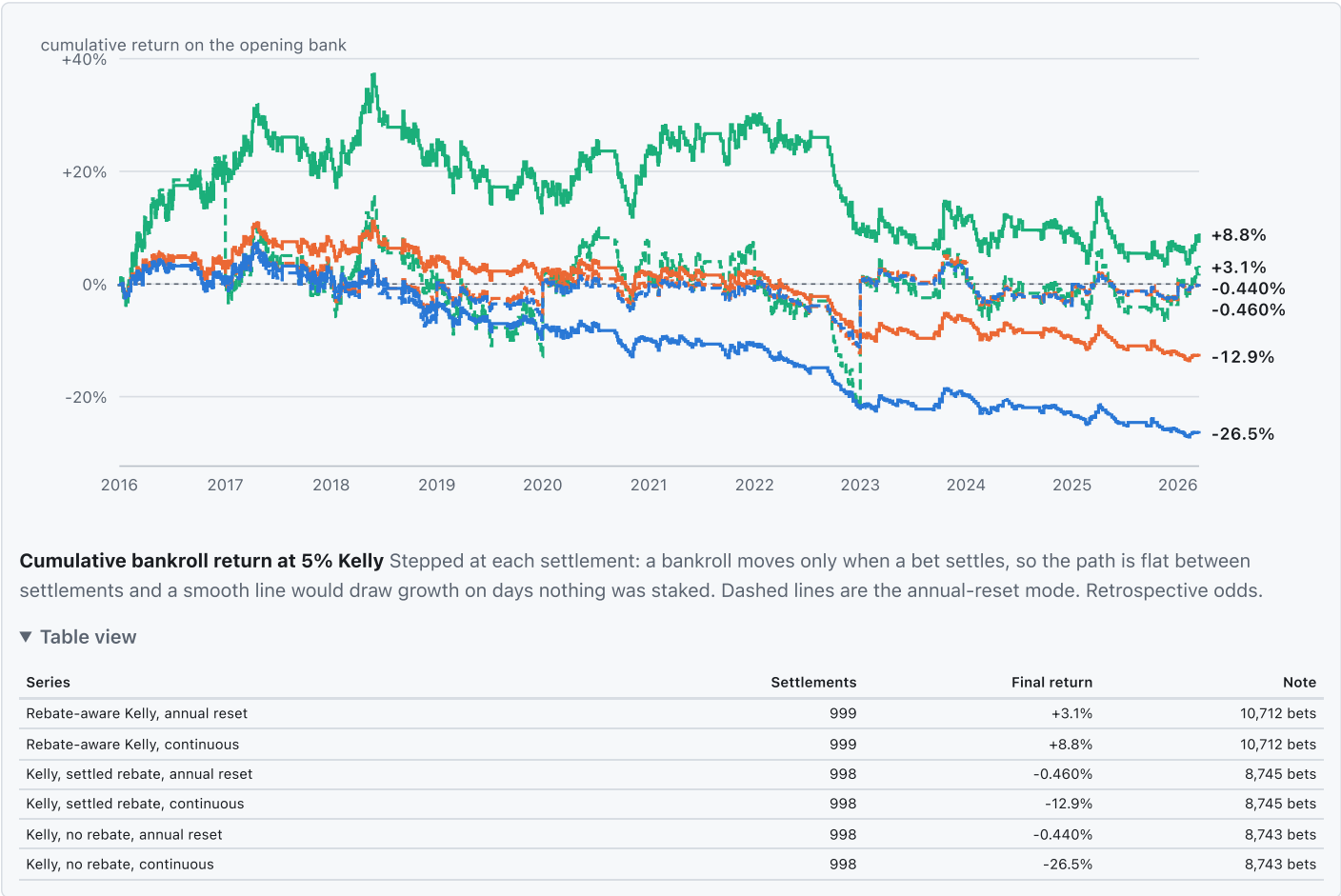
settled bets

REBATE RECEIVED

HK\$8,676,900

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	1,312	1,108	1,108
2017	1,282	1,060	1,060
2018	1,163	972	972
2019	1,217	1,026	1,026
2020	1,239	1,049	1,049
2021	962	742	742
2022	931	751	751
2023	807	621	621
2024	832	664	664
2025	767	607	605
2026	200	145	145

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 1,312 bets in 2016 against 767 in 2025. Kelly, settled rebate: 1,108 bets in 2016 against 607 in 2025. Kelly, no rebate: 1,108 bets in 2016 against 605 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$8,676,900; Rebate-aware Kelly, continuous HK\$8,676,900; Kelly, settled rebate, annual reset HK\$1,839,190; Kelly, settled rebate, continuous HK\$1,839,190. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

HK\$11.0M, 10% KELLY

OPENING BANKROLL

HK\$10,965,342

scenario hkd_10m, solved independently

PRIMARY FINAL RETURN

-13.9%

Rebate-aware Kelly, continuous

BETS

11,236

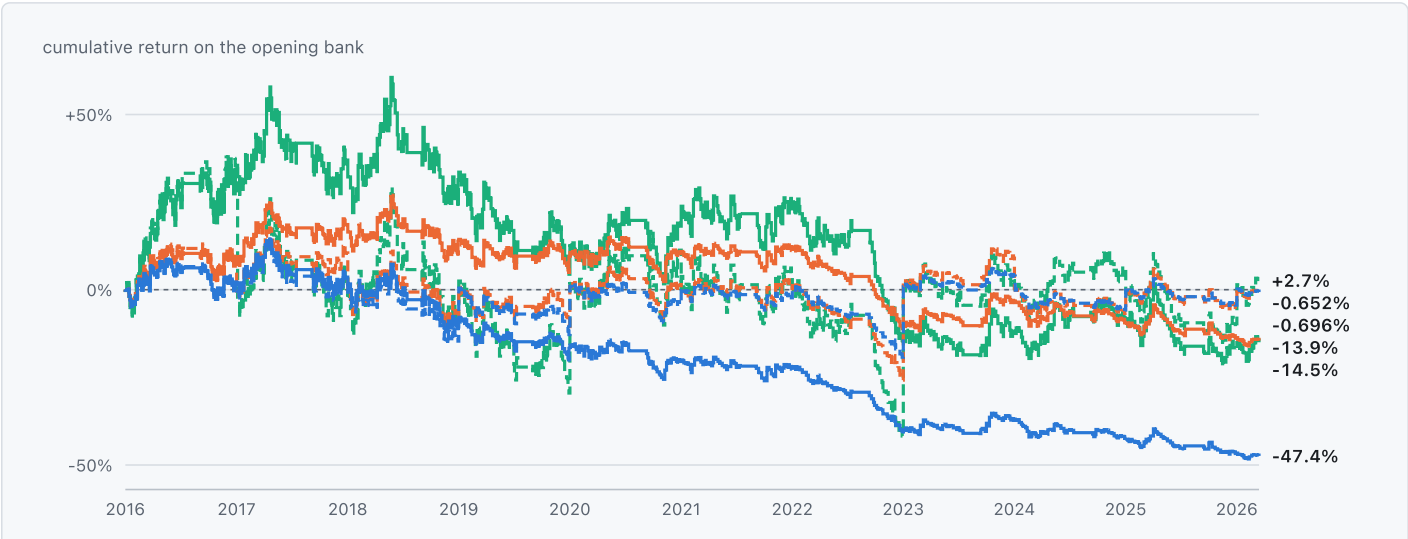
settled bets

REBATE RECEIVED

HK\$17,846,253

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	999	+2.7%	11,236 bets
Rebate-aware Kelly, continuous	999	-13.9%	11,236 bets
Kelly, settled rebate, annual reset	998	-0.696%	8,747 bets
Kelly, settled rebate, continuous	998	-14.5%	8,747 bets
Kelly, no rebate, annual reset	998	-0.652%	8,745 bets
Kelly, no rebate, continuous	998	-47.4%	8,745 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	1,358	1,108	1,108
2017	1,332	1,060	1,060
2018	1,219	972	972
2019	1,254	1,026	1,026
2020	1,281	1,049	1,049
2021	1,009	742	742
2022	989	751	751
2023	853	621	621
2024	892	666	664
2025	825	607	607
2026	224	145	145

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 1,358 bets in 2016 against 825 in 2025. Kelly, settled rebate: 1,108 bets in 2016 against 607 in 2025. Kelly, no rebate: 1,108 bets in 2016 against 607 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$17,846,253; Rebate-aware Kelly, continuous HK\$17,846,253; Kelly, settled rebate, annual reset HK\$5,918,350; Kelly, settled rebate, continuous HK\$5,918,350. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown:

continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

BANKROLL-SCALE SENSITIVITY

Opening bank	Role	Fraction	Final return	Bets	Rebate received
HK\$10.1M	Settlement ablation	1% Kelly	-5.8%	8,695	HK\$0
HK\$10.1M	Sizing reference	1% Kelly	-5.6%	8,695	HK\$17,991
HK\$10.1M	Primary	1% Kelly	-5.7%	9,279	HK\$454,836
HK\$10.1M	Settlement ablation	2% Kelly	-11.3%	8,730	HK\$0
HK\$10.2M	Sizing reference	2% Kelly	-9.4%	8,730	HK\$215,905
HK\$10.4M	Primary	2% Kelly	-2.4%	9,717	HK\$2,329,418
HK\$10.3M	Settlement ablation	5% Kelly	-26.5%	8,743	HK\$0
HK\$10.6M	Sizing reference	5% Kelly	-12.9%	8,745	HK\$1,839,190
HK\$10.6M	Primary	5% Kelly	+8.8%	10,712	HK\$8,676,900
HK\$10.5M	Settlement ablation	10% Kelly	-47.4%	8,745	HK\$0
HK\$11.4M	Sizing reference	10% Kelly	-14.5%	8,747	HK\$5,918,350
HK\$11.0M	Primary	10% Kelly	-13.9%	11,236	HK\$17,846,253
HK\$1.0B	Settlement ablation	1% Kelly	-5.8%	8,751	HK\$0
HK\$1.0B	Sizing reference	1% Kelly	-0.269%	8,751	HK\$58,449,472
HK\$1.0B	Primary	1% Kelly	+0.374%	12,125	HK\$147,494,761
HK\$1.0B	Settlement ablation	2% Kelly	-11.3%	8,751	HK\$0
HK\$1.0B	Sizing reference	2% Kelly	-0.599%	8,751	HK\$120,388,989
HK\$1.0B	Primary	2% Kelly	+0.515%	12,202	HK\$305,675,473
HK\$1.0B	Settlement ablation	5% Kelly	-26.5%	8,751	HK\$0
HK\$1.1B	Sizing reference	5% Kelly	-2.2%	8,751	HK\$324,459,678
HK\$1.1B	Primary	5% Kelly	-0.998%	12,248	HK\$839,974,022
HK\$1.1B	Settlement ablation	10% Kelly	-47.4%	8,751	HK\$0
HK\$1.1B	Sizing reference	10% Kelly	-6.7%	8,751	HK\$724,462,428
HK\$1.1B	Primary	10% Kelly	-9.3%	12,267	HK\$1,894,211,334

Each scenario is solved independently and sequentially by race, because the rebate threshold is a fixed HK\$10,000 and the fraction the solver sees is therefore 1e-3 of a HK\$10M bank and 1e-5 of a HK\$1B one. These are two different optimization problems; the larger bank's stakes are not the smaller bank's scaled up.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,988,826	-0.112%	8,695	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$10,061,598	HK\$9,479,455	-5.8%	8,695	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,989,268	-0.107%	8,695	HK\$17,991
1% Kelly	Kelly, settled rebate, continuous	HK\$10,061,598	HK\$9,496,202	-5.6%	8,695	HK\$17,991
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,971,291	-0.287%	9,279	HK\$454,836
1% Kelly	Rebate-aware Kelly, continuous	HK\$10,113,498	HK\$9,534,776	-5.7%	9,279	HK\$454,836
2% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,979,306	-0.207%	8,730	HK\$0
2% Kelly	Kelly, no rebate, continuous	HK\$10,122,068	HK\$8,975,090	-11.3%	8,730	HK\$0
2% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,978,740	-0.213%	8,730	HK\$215,905
2% Kelly	Kelly, settled rebate, continuous	HK\$10,158,140	HK\$9,199,544	-9.4%	8,730	HK\$215,905
2% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,985,250	-0.148%	9,717	HK\$2,329,418
2% Kelly	Rebate-aware Kelly, continuous	HK\$10,361,374	HK\$10,110,748	-2.4%	9,717	HK\$2,329,418
5% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,955,997	-0.440%	8,743	HK\$0
5% Kelly	Kelly, no rebate, continuous	HK\$10,294,630	HK\$7,562,732	-26.5%	8,743	HK\$0
5% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,953,998	-0.460%	8,745	HK\$1,839,190
5% Kelly	Kelly, settled rebate, continuous	HK\$10,588,522	HK\$9,226,228	-12.9%	8,745	HK\$1,839,190
5% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,305,504	+3.1%	10,712	HK\$8,676,900
5% Kelly	Rebate-aware Kelly, continuous	HK\$10,573,811	HK\$11,504,652	+8.8%	10,712	HK\$8,676,900
10% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,934,828	-0.652%	8,745	HK\$0
10% Kelly	Kelly, no rebate, continuous	HK\$10,548,343	HK\$5,549,689	-47.4%	8,745	HK\$0
10% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,930,397	-0.696%	8,747	HK\$5,918,350
10% Kelly	Kelly, settled rebate, continuous	HK\$11,363,523	HK\$9,712,814	-14.5%	8,747	HK\$5,918,350

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
10% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,267,696	+2.7%	11,236	HK\$17,846,253
10% Kelly	Rebate-aware Kelly, continuous	HK\$10,965,342	HK\$9,437,860	-13.9%	11,236	HK\$17,846,253

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

DAY-BLOCK BOOTSTRAP OF THE BANKROLL

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
flat_stake@edge>=0.02 / block1	2000	HK\$0	HK\$0	0.6975	1.0000	1.0000	0.9120
flat_stake@edge>=0.02 / block3	2000	HK\$0	HK\$0	0.6900	1.0000	1.0000	0.8385
flat_stake@edge>=0.02 / block7	2000	HK\$0	HK\$0	0.7420	1.0000	1.0000	0.8020
kelly_no_rebate@0.01 / block1	2000	HK\$9,467,135	HK\$8,982,747	0.9490	0.0792	0.1151	0.0000
kelly_no_rebate@0.01 / block3	2000	HK\$9,483,568	HK\$8,975,659	0.9475	0.0777	0.1156	0.0000
kelly_no_rebate@0.01 / block7	2000	HK\$9,484,008	HK\$9,039,916	0.9585	0.0755	0.1086	0.0000
kelly_no_rebate@0.02 / block1	2000	HK\$8,952,209	HK\$8,056,628	0.9505	0.1527	0.2176	0.0000
kelly_no_rebate@0.02 / block3	2000	HK\$8,983,571	HK\$8,046,724	0.9500	0.1500	0.2186	0.0000
kelly_no_rebate@0.02 / block7	2000	HK\$8,983,320	HK\$8,162,863	0.9625	0.1459	0.2062	0.0000
kelly_no_rebate@0.05 / block1	2000	HK\$7,516,041	HK\$5,778,908	0.9545	0.3431	0.4616	0.0000
kelly_no_rebate@0.05 / block3	2000	HK\$7,582,232	HK\$5,759,797	0.9555	0.3377	0.4640	0.0000
kelly_no_rebate@0.05 / block7	2000	HK\$7,578,650	HK\$5,968,389	0.9680	0.3295	0.4428	0.0000
kelly_no_rebate@0.1 / block1	2000	HK\$5,482,143	HK\$3,248,416	0.9630	0.5760	0.7170	0.0000
kelly_no_rebate@0.1 / block3	2000	HK\$5,576,816	HK\$3,230,861	0.9640	0.5701	0.7192	0.0000
kelly_no_rebate@0.1 / block7	2000	HK\$5,566,816	HK\$3,458,608	0.9745	0.5587	0.6961	0.0000
kelly_settled_rebate@0.01 / block1	2000	HK\$9,485,057	HK\$8,999,718	0.9460	0.0785	0.1137	0.0000
kelly_settled_rebate@0.01 / block3	2000	HK\$9,499,663	HK\$8,991,303	0.9435	0.0768	0.1141	0.0000
kelly_settled_rebate@0.01 / block7	2000	HK\$9,499,784	HK\$9,056,283	0.9520	0.0744	0.1074	0.0000
kelly_settled_rebate@0.02 / block1	2000	HK\$9,172,385	HK\$8,277,466	0.9020	0.1397	0.2029	0.0000
kelly_settled_rebate@0.02 / block3	2000	HK\$9,208,384	HK\$8,262,431	0.9015	0.1364	0.2017	0.0000
kelly_settled_rebate@0.02 / block7	2000	HK\$9,208,597	HK\$8,382,088	0.9225	0.1326	0.1917	0.0000
kelly_settled_rebate@0.05 / block1	2000	HK\$9,156,492	HK\$7,074,894	0.6880	0.2571	0.3767	0.0000
kelly_settled_rebate@0.05 / block3	2000	HK\$9,247,366	HK\$7,066,128	0.6950	0.2482	0.3722	0.0000
kelly_settled_rebate@0.05 / block7	2000	HK\$9,255,648	HK\$7,324,391	0.7115	0.2421	0.3569	0.0000
kelly_settled_rebate@0.1 / block1	2000	HK\$9,570,223	HK\$5,739,451	0.5525	0.4017	0.5690	0.0000
kelly_settled_rebate@0.1 / block3	2000	HK\$9,751,429	HK\$5,726,310	0.5300	0.3866	0.5598	0.0000
kelly_settled_rebate@0.1 / block7	2000	HK\$9,777,694	HK\$6,124,365	0.5385	0.3796	0.5439	0.0000
rebate_aware_kelly@0.01 / block1	2000	HK\$9,528,624	HK\$8,823,201	0.8440	0.0936	0.1420	0.0000
rebate_aware_kelly@0.01 / block3	2000	HK\$9,538,445	HK\$8,825,473	0.8440	0.0926	0.1423	0.0000
rebate_aware_kelly@0.01 / block7	2000	HK\$9,537,270	HK\$8,913,712	0.8810	0.0904	0.1365	0.0000
rebate_aware_kelly@0.02 / block1	2000	HK\$10,059,725	HK\$8,391,870	0.4800	0.1515	0.2483	0.0000
rebate_aware_kelly@0.02 / block3	2000	HK\$10,085,282	HK\$8,400,287	0.4595	0.1464	0.2426	0.0000
rebate_aware_kelly@0.02 / block7	2000	HK\$10,136,782	HK\$8,443,814	0.4485	0.1456	0.2380	0.0000
rebate_aware_kelly@0.05 / block1	2000	HK\$11,505,428	HK\$7,199,074	0.3195	0.3208	0.4930	0.0000
rebate_aware_kelly@0.05 / block3	2000	HK\$11,547,287	HK\$7,132,201	0.3130	0.3117	0.4865	0.0000
rebate_aware_kelly@0.05 / block7	2000	HK\$11,491,614	HK\$7,374,477	0.2930	0.3061	0.4733	0.0000
rebate_aware_kelly@0.1 / block1	2000	HK\$9,396,673	HK\$3,736,990	0.5360	0.5908	0.7716	0.0000
rebate_aware_kelly@0.1 / block3	2000	HK\$9,389,019	HK\$3,771,785	0.5440	0.5711	0.7738	0.0000
rebate_aware_kelly@0.1 / block7	2000	HK\$9,559,259	HK\$3,985,122	0.5340	0.5684	0.7584	0.0000

Race days resampled with replacement, so results from one meeting stay together. This is sampling risk in the settled path, not model risk. 39 of 234 settled series shown, filtered to model view softmax_bagged.

MODEL-ASSUMED MONTE CARLO

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
flat_stake@edge>=0.02	1000	HK\$0	HK\$0	1.0000	1.0000	1.0000	1.0000
kelly_no_rebate@0.01	1000	HK\$11,388,253	HK\$10,706,077	0.0020	0.0186	0.0309	0.0000
kelly_no_rebate@0.02	1000	HK\$12,985,215	HK\$11,430,699	0.0000	0.0367	0.0604	0.0000
kelly_no_rebate@0.05	1000	HK\$19,144,977	HK\$13,783,805	0.0000	0.0873	0.1412	0.0000
kelly_no_rebate@0.1	1000	HK\$35,498,066	HK\$18,350,579	0.0000	0.1695	0.2638	0.0000
kelly_settled_rebate@0.01	1000	HK\$11,388,451	HK\$10,707,550	0.0020	0.0186	0.0309	0.0000
kelly_settled_rebate@0.02	1000	HK\$12,986,319	HK\$11,425,539	0.0000	0.0367	0.0604	0.0000
kelly_settled_rebate@0.05	1000	HK\$19,231,815	HK\$13,808,116	0.0000	0.0878	0.1400	0.0000
kelly_settled_rebate@0.1	1000	HK\$35,156,136	HK\$18,888,267	0.0010	0.1688	0.2630	0.0000

Winners drawn from the model's own probabilities. 9 of 53 settled series shown, filtered to model view softmax_bagged.

One simulation was declined rather than approximated

Refusals are recorded rather than omitted, so a reader can see the strategy was considered and declined. Rebate-aware sizing cannot reuse stale bankroll-dependent fractions on a divergent path.

The realised path, the day-block bootstrap and the model-assumed MC answer different questions and must not be shown as one uncertainty band.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix and provenance

CONFIGURATION

Item	Value
Features	28
Features the booster learns on	28
Opening bankroll	10,000,000
Bankroll scenario	hkd_10m
Scenario role	primary
Rebate threshold as a bankroll fraction	0.001000

definition, raw_source, units, default_behavior, higher_value_meaning and known_caveats are intentionally blank. They are claims about the domain, not properties derivable from the model, and inventing them would produce documentation that reads as authoritative and is not.

PROVENANCE

Item	Value
Source commit	eb0e98021dda
Source tree	dirty at generation
Config digest	sha256:73bf1d82bb8d73784ab90d0e081755cfe5c3d44b17687f1be29ff951c27591ad
Identity source	artifact
Model view	softmax_bagged
Artifact root	results/softmax_report
Inputs read	18 artifact(s), locked by digest in run_lock.json
Evaluation race keys	7933 races, sha256:342d95ee674808a0d7ae4d0df5b0f958da1898863e25c2898bb71d9f58392e58
Environment: numpy	2.5.2
Environment: pandas	3.0.5

Item	Value
Environment: python	3.12.11
Environment: xgboost	3.4.1

The generating source tree was dirty

This run's provenance records uncommitted changes, so the commit named above does not fully describe the code that produced these numbers. Treat the figures as indicative until the report is regenerated from a clean tree.

Every input this report read is listed in `run_lock.json` by content digest, together with the lock digest of the run that produced them -- which is what chains a published figure back through the report to the fit and from there to the feature table's own digest.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Softmax against the market on the same races. It does not measure this model against the other published models except through that shared yardstick.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.

Unverified source state

This run recorded a dirty source tree, so its numbers cannot be reproduced from the named commit alone.